

# THE PILLOW FIELDS · EMPEROR BEAR · EVERY UNIT, EVERY DIFFICULTY

What every Snugg, cub, paratrooper, guard, Bluecoat, cannon, cavalryman and Emperor Bear does to your men on EASY, MEDIUM, HARD and MEGA, with every body's HP and head-count · read out of the live game v32.45 on Nov 14, 2026 14:06 EST · every damage number checked on the game's own machines (Chrome 349/349 · WebKit 349/349) · for Raziya

**The one thing to know:** in the Pillow Fields **no blow is an instant KO any more** (CARD-BDC2): the cub's pounce, the paratrooper's landing, a Bluecoat's sword, the hooves, the saddle swing and the Emperor's own sword each take a **priced bite of 36 HP (49.5 on MEGA)**, landing whole through the man's bar ÷ his HP. A rifleman at the lab's levels still falls to one; **every other man stands on at least 15 % of a full bar** (CARD-RZB) and falls to the next; the heavies take a fraction (a level-50 flame 2.08 of 7 bar hits); every hero stands. A Snugg's or a paratrooper's **glob** (2 bar hits, landing at 50 % by the damper) and the cannon's **lob** (0.9 × a rifleman's HP, whole: CARD-EBA) wear a man down. **Above 80 % of his first bar the Emperor walks and cuts from the saddle**, one man every 1.5 s; his unhorse takes 3 s and pays 150 dog tags (CARD-EBP).

THE BOSS LAB fights the Emperor with its stock army at **LEVEL 50 on EASY, 57 on MEDIUM, 63 on HARD and 70 on MEGA**: Sarge with the saber, the jeep, the tank, the dogfighter and the bomber, and seven men at that rank (rifleman, M. gunner, bazooka, mortar, medic, flame, sweeper). The radio man, the grenadier, the commando and the spy are a tap away in THE PLATOON. Every "level" number on this page is that army at that tier's level.

## What changed recently (since Oct 29; the newest first)

| live since | when | card | what it does in this fight |
|---|---|---|---|
| v31.25 | Nov 10 | **CARD-EBP · THE EMPEROR'S TWO PHASES** | Above 80 % of his first bar he walks and cuts from the saddle, one man every 1.5 s inside 110 px (the hooves and the charge come under it). The unhorse takes 3 s: the scamper, his line, the whistle, every wave held till then (it was the 0.8 s keel and EN GARDE!); 150 dog tags for phase one; his parry in phase two (20 % of a saber or Claymore Sarge's swings); no wave counter in a park boss fight. |
| v31.20 | Nov 10 15:27 | **CARD-EBA · THE EMPEROR'S ARMY** | A cannon's lob lands 0.9 × a rifleman's HP at your man's rank through his bar, whole (a level-50 rifleman 6.3 of his 7 bar hits, a flame 1.35; it was half the bar × the share × the damper: 1.62 on that rifleman), and it hurts the bears in its splash too (a Bluecoat a fifth, the Emperor two fifths). A Bluecoat's cut on a Flamer costs 3.6 HP (34 cuts on a level-50 flame of 121 HP). A jeep that rams a Bluecoat takes a smaller dent; an officer at full health cuts the jeep down. The paratroopers come down later: at half his first bar or 60 s in, and again 12 s after the unhorse. |
| v31.00 | Nov 6 | **CARD-RZB · THE FLOOR, THE CAP** | No single hit fells a man other than a rifleman from a full bar: the bite that would leaves him on 15 % of it (the medic, the spy, a pistol or saber Sarge at these levels), the next fells him. One blow reaches at most 3 · 4 · 4 · 5 men on EASY · MEDIUM · HARD · MEGA (measured: one landing over two men more than that bit exactly that many). The jeep pays for what it runs down (a dent and a slowing). |
| v29.72 | Nov 4 11:12 | **CARD-KHP · THE KINGS FROM RANK 30** | King Yeti, King Rex, King Croak and the Emperor Bear hero stand 120 more HP and bar hits from their own rank 30: at level 50 King Rex has 296 HP and 225 bar hits (was 176 and 105), King Croak 321 and 285 (was 201 and 165); Jane and Princess Croak as they were. A glob still lands a whole bar hit on them; a lob and a priced bite take a little less of their bar. |

| live since | when | card | what it does in this fight |
|---|---|---|---|
| v29.70 | Nov 3 | **CARD-BDC2 · NO INSTANT KO** | A blow that KO'd a man outright (the cub's pounce, the paratrooper's landing, a Bluecoat's and the officer's cut, the hooves, the saddle swing, the Emperor's sword) is a priced bite now: 0.9 × the machine gunner's 40 HP at rank 57 (the fight's level 100; MEGA level 140, rank 80) = **36 HP (49.5 on MEGA)**, landing whole through his bar ÷ his HP. On EASY the cub's pounce now fells 1 of the 7 stock men (it felled 6: the v29.45 read); the Emperor's three cuts fell one man of three (they felled 3); every hero stands. THE HEAVY CLASS: the commando, the flame and the bazooka stand at least twice the machine gunner's HP (the bazooka 80 HP at level 50, twice the machine gunner's 40). |
| v29.41 | Nov 2 18:11 | **CARD-DHP · THE BONUS DOUBLED, THE SHOTGUN'S HEALTH** | The bazooka's and the commando's tenth-rank HP bonus doubled (+5 more HP a tenth rank for the bazooka, +5 for the commando): at the lab's levels 50 · 57 · 63 · 70 the bazooka has 74 · 74 · 88 · 101 HP (was 49 · 49 · 58 · 66), the commando 83 · 83 · 96 · 110 (was 58 · 58 · 66 · 75). Shotgun Sarge +20 HP for every ten shotgun ranks while the shotgun is in his hands: 135 · 135 · 157 · 180 HP (was 35 · 35 · 37 · 40); his other irons as they were. A glob or a lob lands a hair lighter on them (the share); the KO strikes do not care. |
| v29.40 | Nov 2 18:03 | **CARD-FLT · THE FLAME'S RUNGS TRIPLED** | The flame's tenth-rank HP rungs through level 50 tripled: +60 HP at every lab level, 121 · 121 · 130 · 140 (was 61 · 61 · 70 · 80); every other man as he was. |
| v29.37 | Nov 2 16:49 | **CARD-RXT · THE TIME-TO-KILL CHART** | The designers' time-to-kill chart sets the Emperor's bar for one commander on EASY, MEDIUM and HARD: 1,240 · 1,654 · 2,481 at rung 50 or under, 1,240 · 1,653 · 2,478 at the lab's levels 50 · 57 · 63 (was 152 · 233 · 287 at the lab's levels, 152 · 205 · 228 at rung 50 or under). The price of the player's rung no longer moves his bar (the player-damage-threshold pass cancelled it first at v29.30, its multiplier figured over the priced bar; the chart wrote its own numbers over that at v29.37), so the price cancels and the lab's level lands 0 to 3 HP under the chart's number. The cavalryman's bar rides the same multiplier: 514 · 605 · 820 (was 63 · 85 · 95). A party of two: 2,358 · 3,147 · 4,712 (was 289 · 390 · 433). MEGA keeps the player-damage-threshold pass's table (the row below). |
| v29.30 | Nov 2 15:45 | **CARD-PDT · THE PLAYER DAMAGE THRESHOLD** | The pass writes the boss's bar for one commander into his row's multiplier at four decimals, figured over the solo bar with the rung's price in it: the price cancels on the Emperor's bar and stays on his Bluecoats, officer and cannons; the cavalryman's boss-typed row takes the same multiplier, so a solo player's price now LOWERS his bar (605 at the lab's level 57 against 687 at rung 50 or under on MEDIUM) while a party's rows carry no price. On MEGA the Emperor's bar is the pass's table: 6,345 at rung 50 or under, 6,339 at the lab's level 70 (was 390 at the lab's level, 274 at rung 50 or under); the cavalryman 1,845 (was 113); a party of two 12,042 (was 520). The chart (the row above) took EASY, MEDIUM and HARD from this table the same day. |
| v29.17 | Nov 1 23:52 | **CARD-DMP · THE DAMPER** | Every enemy blow on your men lands at 50 % in the campaign and the boss fights (the damper takes 50 % off on EASY, MEDIUM, HARD and MEGA alike). Here a glob lands 0.93 of a level-50 rifleman's 7 bar hits (it was 1.85), a lob a quarter of his bar (the lob of CARD-EBA replaced it on Nov 10), a Bluecoat's cut or the saddle swing on Sarge or a hero, and the hooves on a hero, 0.5 bar hit (× his share on Sarge): a man stands twice the blows he did. A KO is not dampened: the pounce, the landing, the swords and the hooves on a man land as shipped. |
| v28.93 | Oct 31 22:56 | **CARD-KCF · KING CROAK** | King Croak the hero is tougher on every rank: 321 HP and 285 bar hits at level 50 (the heroes' table). Every blow lands on him as on any hero. |
| v28.74 | Oct 31 00:11 | **CARD-HP3 · THE HEAVIES** | +3 HP every tenth rank for the bazooka, grenadier, flame and commando. A glob or a lob lands a hair lighter on them (the share); the KO strikes do not care. |
| v28.68 | Oct 30 23:31 | **CARD-BRB · THE BOMBER** | Your bomber flies in every boss fight now, the Pillow Fields included, and aims at the thickest knot of plush (or the Emperor). |
| v28.65 | Oct 30 19:39 | **CARD-LRB · ONE HP NUMBER** | Your man's HP is the number on his unit sheet. A glob, a lob and a Bluecoat's cut on Sarge still count in bar hits, but land a little lighter on tough men: 90 % + 10 % × bar ÷ HP (a level-50 flame takes 1.81 of a glob's 2, a rifleman 1.85, before the damper). |
| v28.46 | Oct 30 12:57 | **CARD-BLP · THE BOSS LAB'S STOCK ARMY** | The lab fights the Emperor with its own army at LEVEL 50 · 57 · 63 · 70 (the lab's level sheet for the fields reads 50–70, spread over the four tiers): the one this page reads. Past rung 50 that level's price rode his bars until the player-damage-threshold pass (v29.30; the chart then wrote its own numbers over it at v29.37); today it rides his Bluecoats, officer and cannons (×1 · 1.136 · 1.257 · 1.422) and lowers the cavalryman's bar. |
| read | Oct 10 | **CARD-BHP · ONE BOSS, MORE HEALTH** | A party does not multiply the Emperor: it grows his bars (×1 + 0.9 a commander past the first); the cavalry's bar grows the same way (the table below). |

| live since | when | card | what it does in this fight |
|---|---|---|---|
| — | — | **UNCHANGED SINCE THEN** | The globs of the Snugg and the paratrooper, a Bluecoat's cut on Sarge or a hero, the hunt on foot, the royal funnel and the battery, the bars, the wave sheets. The stone wall and the reflect are switched off (0904). Bear Park's riders and the box (CARD-BPB, Nov 13) are Bear Park's alone; the fields' own toy box card (CARD-EBX) is not in the game. The versions' dates above are the game's own git log where it was read (the Big Mac, Nov 2); CARD-EBA landed as v31.20 (Nov 10 15:27 EST), CARD-EBP as v31.25 (the git log on the Big Mac). |

## How a hit is counted

Your men carry **HP**: the number on their unit sheet (a level-50 flame has 121, a rifleman 26; at level 70: 140 and 33). Their health bar is also cut into a few **bar hits** (7 for most men at these levels, 6 for the medic and spy, 8 for the commando, 12 for the saber Sarge). Three kinds of hit in this fight:

- **Priced bites** (the cub's pounce, the paratrooper's landing, a Bluecoat's cut on a man, the hooves, the Emperor's sword from the saddle and on foot, the saddle swing; since CARD-BDC2 v29.70 no hit is an instant KO): 36 HP (49.5 on MEGA) off his HP, landing whole: bar hits = that HP × his bar ÷ his HP (a level-50 rifleman 9.69 of his 7: he falls; a flame 2.08). A bite that would empty a full bar leaves a man other than a rifleman on 15 % of it (CARD-RZB, v31.00); the next one fells him. One blow reaches at most 3 · 4 · 4 · 5 men on EASY · MEDIUM · HARD · MEGA.
- **Bar hits** (a glob: 2; a Bluecoat's cut on Sarge or a hero: 1; the Emperor's or the cavalry's blow on a hero: 1): since v28.65 a bar hit lands 90 % + 10 % × bar ÷ HP, and the damper lets 50 % of that land, so a level-50 rifleman takes 0.93 of his 7 from a glob (3.4 HP of 26). A man who takes no hit for 4 s mends one bar hit, and again every 4 s.
- **The cotton lob** (CARD-EBA): 0.9 × a rifleman's HP at the man's rank, through his bar ÷ his HP, landing whole, so 6.3 of a level-50 rifleman's 7: 2–6 lobs 3.5 s apart KO every man (4–14 KO a hero); the jeep and the tank take one hull hit a lob.
- **THE DAMPER** (since v29.17): in the campaign and the boss fights a glob, a Bluecoat's cut on Sarge or a hero and a hero's blow under the hooves land at **50 %** (the game takes 50 % off on EASY, MEDIUM, HARD and MEGA alike, read in each tier's lab fight); a priced bite and a lob land whole.

Saves still roll on every hit: a jetting, dropping or dragged man cannot be touched, the pet shield, the deflect dome, the absorbs, and **the saber Sarge's block turns a glob, a lob or a Bluecoat's cut 50 % of the time**; a pounce rolls it twice (on the strike, then on the priced bite: about 75 %, measured 32, 29, 32, 34 of 40 turned on the four tiers on Chrome, 35, 28, 31, 33 on WebKit). **The Emperor's cut and a moving horse come already rolled**: only the bite's own roll is left (measured: 154 of the king's cuts turned in 80 exchanges, and no Sarge fell to the first cut that landed). The Claymore Sarge's counted block and the riot plate keep their say. A Bluecoat whose man is a saber Sarge mid-charge waits for the strike to land, then cuts.

## The strikes · what a hit does by difficulty

| hit | EASY | MEDIUM | HARD | MEGA | who it hits | since |
|---|---|---|---|---|---|---|
| **THE CUB'S POUNCE** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | the man he hops on, then the next within 150 px: a priced bite (the HP × his bar ÷ his HP, landing whole); from a full bar a man other than a rifleman stands on 15 % | v29.70 CARD-BDC2: no instant KO (it was a KO) · v31.00 CARD-RZB: the floor |
| **THE SNUGG'S GLOB · bar hits** | **2 × 0.5 × share** | **2 × 0.5 × share** | **2 × 0.5 × share** | **2 × 0.5 × share** | one man within 280 px, every 3.6 s | v29.17: × the damper (v28.65: lands 90 % + 10 % × bar ÷ HP of that) |
| **THE PARATROOPER'S THUMP** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | every man within 76 px of his landing (the shadow sways up to 9 px), at most 3 · 4 · 4 · 5 men a landing | v29.70 CARD-BDC2 (it was a KO) · v31.00 CARD-RZB: the floor and the cap |
| **THE PARATROOPER'S GLOB · bar hits** | **2 × 0.5 × share** | **2 × 0.5 × share** | **2 × 0.5 × share** | **2 × 0.5 × share** | one man within 230 px, every 2.6 s | v29.17: × the damper |
| **A BLUECOAT'S CUT** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | the nearest man within 74 px of each cut, three cuts a lunge; on a Flamer 3.6 HP (CARD-EBA) | v29.70 CARD-BDC2 (it was a KO) · Nov 10 CARD-EBA: the Flamer |
| **A BLUECOAT'S CUT · Sarge, a hero** | **1 × 0.5 × share** | **1 × 0.5 × share** | **1 × 0.5 × share** | **1 × 0.5 × share** | a bar hit through his saves (the saber turns half) | v29.17: × the damper |
| **THE COTTON LOB · bar hits** | **0.9 × a rifleman's HP** | **0.9 × a rifleman's HP** | **0.9 × a rifleman's HP** | **0.9 × a rifleman's HP** | every man within 50 px of the burst, through his bar ÷ his HP, landing whole (a rifleman 0.9 of his bar); the jeep: a hull hit; the bears in the splash take it too (a Bluecoat a fifth, the Emperor two fifths) | Nov 10 CARD-EBA (it was half the bar × the share × the damper) |
| **THE HOOVES (Emperor · cavalry)** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | a man within 52 px of a moving horse; Sarge's and Jane's fall is priced too; a hero 0.5 bar hit + the throw a pass (the Emperor's hooves land it every pass, the cavalry never a hero's last); the Emperor rides them under 80 % of his first bar | v29.70 CARD-BDC2 (it was a KO; Sarge and Jane fell) · CARD-EBP: under 80 % |
| **THE EMPEROR'S SADDLE SWING (at rest, under 80 %)** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | a man within 52 px every 0.6 s: a priced bite; a Sarge with a blade in hand: 1 bar hit × 0.5 × his share and keeps his feet; a gun Sarge, a hero: the same blow + thrown 135–185 px (the pistol Sarge 182–185, King Rex 135; measured) | v29.70 CARD-BDC2: on a man a priced bite (it was a KO) · v29.17: Sarge's and a hero's blow × the damper |
| **THE EMPEROR'S SWORD (on foot)** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | three cuts a lunge, each the nearest uncut man within 74 px: a priced bite each; on Sarge the fall comes rolled and priced (his blade turns the bite in its own roll); his parry in phase 2: 20 % of a saber or Claymore Sarge's swings, a 0.9 s stall | v29.70 CARD-BDC2 (it was a KO, three men down) · CARD-EBP: the parry |

| hit | EASY | MEDIUM | HARD | MEGA | who it hits | since |
|---|---|---|---|---|---|---|
| **THE SWORD FROM THE SADDLE (above 80 %)** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | at or above 80 % of his first bar he walks his horse (70 px/s) at the nearest man and cuts him every 1.5 s inside 110 px: one man a swing, a priced bite (measured) | CARD-EBP (Nov 10): new |
| **BODY HP · the plush** | **×1** | **×1.35** | **×1.5** | **×1.8** | the guards, Snuggs, paratroopers, cubs and cavalry: the tier alone | unchanged |
| **BODY HP · his Bluecoats, officer and cannons** | **×1 × 1** | **×1.35 × 1.136** | **×1.5 × 1.257** | **×1.8 × 1.422** | the tier × THE PRICE of the player's rung past 50 (×1 · 1.136 · 1.257 · 1.422 solo at the lab's levels 50 · 57 · 63 · 70; ×1 in a party or at rung 50 or under): the funnel's bodies keep the price the Emperor's bar no longer carries | unchanged |
| **HEAD-COUNTS** | **×1** | **×1.25** | **×1.5** | **×1.5** | the Snuggs and the paratroopers; the guards, the cubs and the cavalry stay 4 · 3 · 3 | unchanged |
| **THE EMPEROR'S BAR (lab's level, one commander)** | **1,240 × 2 lives** | **1,653 × 2 lives** | **2,478 × 2 lives** | **6,339 × 2 lives** | the designers' time-to-kill chart's bar for one commander on EASY, MEDIUM and HARD, the player-damage-threshold pass's table on MEGA (1,240 · 1,654 · 2,481 · 6,345 at rung 50 or under), written into his row's multiplier at four decimals over the priced solo bar: the price of the rung cancels, and the lab's levels 50 · 57 · 63 · 70 land 0 to 6 HP under the chart's number | v29.37 (the chart) · v29.30 (the pass: MEGA) |
| **A CAVALRYMAN'S BAR (lab's level, one commander)** | **514** | **605** | **820** | **1,845** | his row takes the Emperor's multiplier, so his bar rides the Emperor's; figured over the priced bar, a solo player's price LOWERS it: 514 · 687 · 1,032 · 2,626 at rung 50 or under | v29.37 (the chart) · v29.30 (the pass) |

share = 90 % + 10 % × the man's bar hits ÷ his HP (CARD-LRB); × 0.5 = THE DAMPER (CARD-DMP): what lands of an enemy's blow on your men, read in each tier's lab fight. The fields' dials sit in THE BOSS LAB's RULES under THE PILLOW FIELDS: a typed cell is what the next lab fight uses.

## Who survives · the lab's levels 50 · 57 · 63 · 70

How many of that hit in a row KO a fresh man at the tier's own level. **KO** = one hit (a priced bite that empties a rifleman's bar). Hover a cell for the numbers.

KO one hit · 2 two · 3 three · 4 four · 5+ five or more

· a dot in the corner = measured on the game's own machine in this read · ½ = the saber turns half · ¾ = his blade turns about 3 pounces in 4 · KO·2 = the bites in a row by tier when they differ (EASY · MEDIUM · HARD · MEGA)

| | HP / BAR HITS AT THE LAB'S LEVEL | | | | GLOBS TO KO (Snugg, paratrooper) | | | | CANNON LOBS TO KO | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| man | EASY 50 | MEDI 57 | HARD 63 | MEGA 70 | E | M | H | G | E | M | H | G | BLUECOAT'S CUT | POUNCE · THUMP | HOOVES · SWORD |
| THE LAB'S STOCK ARMY | | | | | | | | | | | | | | | |
| **RIFLEMAN** | **26** / 7 | **26** / 7 | **29** / 7 | **33** / 7 | 8 | 8 | 8 | 8 | 2 | 2 | 2 | 2 | KO | KO | KO |
| **M. GUNNER** | **40** / 7 | **40** / 7 | **45** / 7 | **50** / 7 | 8 | 8 | 8 | 8 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **BAZOOKA** | **80** / 7 | **80** / 7 | **90** / 7 | **101** / 7 | 8 | 8 | 8 | 8 | 4 | 4 | 4 | 4 | 3 | 3 | 3 |

| | HP / BAR HITS AT THE LAB'S LEVEL | | | | GLOBS TO KO (Snugg, paratrooper) | | | | CANNON LOBS TO KO | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| man | EASY 50 | MEDI 57 | HARD 63 | MEGA 70 | E | M | H | G | E | M | H | G | BLUECOAT'S CUT | POUNCE · THUMP | HOOVES · SWORD |
| **MORTAR** | 23 / 7 | 23 / 7 | 26 / 7 | 30 / 7 | 8 | 8 | 8 | 8 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **MEDIC** | 22 / 6 | 22 / 6 | 26 / 6 | 29 / 6 | 7 | 7 | 7 | 7 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **FLAME** | 121 / 7 | 121 / 7 | 130 / 7 | 140 / 7 | 8 | 8 | 8 | 8 | 6 | 6 | 5 | 5 | 34·34·37·39 | 4·4·4·3 | 4·4·4·3 |
| **SWEEPER** | 24 / 7 | 24 / 7 | 28 / 7 | 31 / 7 | 8 | 8 | 8 | 8 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **SARGE · SABER** | 35 / 12 | 35 / 12 | 37 / 12 | 40 / 12 | 13 | 13 | 13 | 13 | 2 | 2 | 2 | 2 | 26 | 2 ¾ | 2 |
| A TAP AWAY (THE PLATOON, SARGE'S OTHER IRONS) | | | | | | | | | | | | | | | |
| **RADIO MAN** | 24 / 7 | 24 / 7 | 28 / 7 | 31 / 7 | 8 | 8 | 8 | 8 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **GRENADIER** | 45 / 7 | 45 / 7 | 53 / 7 | 61 / 7 | 8 | 8 | 8 | 8 | 2 | 2 | 3 | 3 | 2 | 2 | 2 |
| **COMMANDO** | 83 / 8 | 83 / 8 | 96 / 8 | 110 / 8 | 9 | 9 | 9 | 9 | 4 | 4 | 4 | 4 | 3 | 3 | 3 |
| **SPY** | 27 / 6 | 27 / 6 | 32 / 6 | 36 / 6 | 7 | 7 | 7 | 7 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **SARGE · PISTOL** | 35 / 11 | 35 / 11 | 38 / 11 | 41 / 11 | 12 | 12 | 12 | 12 | 2 | 2 | 2 | 2 | 24 | 2 | 2 |
| **SARGE · SHOTGUN** | 135 / 8 | 135 / 8 | 157 / 8 | 180 / 8 | 9 | 9 | 9 | 9 | 6 | 6 | 7 | 7 | 18 | 4·4·5·4 | 4·4·5·4 |
| **SARGE · SNIPER** | 26 / 8 | 26 / 8 | 28 / 8 | 31 / 8 | 9 | 9 | 9 | 9 | 2 | 2 | 2 | 2 | 18 | 2 | 2 |
| **SARGE · CLAYMORE** | 68 / 23 | 70 / 23 | 71 / 23 | 71 / 23 | 25 | 25 | 25 | 25 | 3 | 3 | 3 | 3 | 50 | 2* | 2 |

GLOBS: by the sum of the hits; a Snugg's globs come 3.58–4.17 s apart and a man mends one bar hit for every 4 s he is left alone, and since the damper a glob lands less than one bar hit, so every mend undoes about a whole glob: a lone Snugg needs more than the sum (measured 8–23 globs on 7-bar men over 40 runs on both engines, 8 by the sum; 151 of the 490 gaps passed the 4 s mend, 1 run needed 23). The paratrooper's globs come 2.58–3.17 s apart, inside the mend. LOBS: 3.5 s apart, inside the mend; each lands 0.9 × a rifleman's HP at the man's rank through his bar, whole (CARD-EBA). BLUECOAT'S CUT on Sarge: 1 bar hit × the damper's 50 % through his saves. BLUECOAT'S CUT on a man, POUNCE · THUMP, HOOVES · SWORD: the priced bites in a row to a KO (CARD-BDC2's 36 HP (49.5 on MEGA), landing whole; on a Flamer a Bluecoat's cut is CARD-EBA's 3.6 HP); a bite that would empty a full bar leaves a man other than a rifleman on 15 % (CARD-RZB), so he needs two; one number when the four tiers agree, else EASY · MEDIUM · HARD · MEGA. * The Claymore Sarge's own block can eat a bite whole (not measured here).

## The heroes at the lab's level (one in command), measured

| hero | HP (E · M · H · G) | bar hits | pounce · thump | a glob | a cotton lob | the cavalry's hooves | the Emperor's sword |
|---|---|---|---|---|---|---|---|
| KING YETI | **242 · 244 · 251 · 258** | 234 · 234 · 240 · 246 | 34.81 · 34.52 · 34.42 · 47.2 bar hits a bite, stands | 1 · 1 · 1 · 1 bar hits | 22.63 · 22.44 · 24.96 · 28.32 of 234 · 234 · 240 · 246 (0.9 × a rifleman HP at his rank, whole) | 0.5 bar hit + thrown 154–160 px a pass | 34.81 · 34.52 · 34.42 · 47.2 bar hits a cut |
| KING REX | **296 · 299 · 320 · 329** | 225 · 225 · 251 · 257 | 27.36 · 27.09 · 28.24 · 38.67 bar hits a bite, stands | 1 · 1 · 1 · 1 bar hits | 17.79 · 17.61 · 20.47 · 23.2 of 225 · 225 · 251 · 257 (0.9 × a rifleman HP at his rank, whole) | 0.5 bar hit + thrown 158–219 px a pass | 27.36 · 27.09 · 28.24 · 38.67 bar hits a cut |
| KING CROAK | **321 · 323 · 330 · 337** | 285 · 285 · 291 · 297 | 31.96 · 31.76 · 31.75 · 43.62 bar hits a bite, stands | 1 · 1 · 1 · 1 bar hits | 20.78 · 20.65 · 23.02 · 26.17 of 285 · 285 · 291 · 297 (0.9 × a rifleman HP at his rank, whole) | 0.5 bar hit + thrown 23–29 px a pass | 31.96 · 31.76 · 31.75 · 43.62 bar hits a cut |
| JANE | **108 · 111 · 118 · 126** | 56 · 56 · 62 · 68 | 18.67 · 18.16 · 18.92 · 26.71 bar hits a bite, stands | 1 · 1 · 1 · 1 bar hits | 12.13 · 11.81 · 13.71 · 16.03 of 56 · 56 · 62 · 68 (0.9 × a rifleman HP at his rank, whole) | her fall is priced: 18.67 · 18.16 · 18.92 · 26.71 bar hits a pass, she stands | 18.67 · 18.16 · 18.92 · 26.71 bar hits a cut |
| EMPEROR BEAR (the hero) | **304 · 308 · 329 · 338** | 225 · 225 · 251 · 257 | 26.64 · 26.3 · 27.46 · 37.64 bar hits a bite, stands | 1 · 1 · 1 · 1 bar hits | 17.32 · 17.09 · 19.91 · 22.58 of 225 · 225 · 251 · 257 (0.9 × a rifleman HP at his rank, whole) | 0.5 bar hit + thrown 89–100 px a pass | 26.64 · 26.3 · 27.46 · 37.64 bar hits a cut |

| hero | HP (E · M · H · G) | bar hits | pounce · thump | a glob | a cotton lob | the cavalry's hooves | the Emperor's sword |
|---|---|---|---|---|---|---|---|
| PRINCESS CROAK | **82 · 83 · 90 · 97** | 65 · 65 · 71 · 77 | 28.54 · 28.19 · 28.4 · 39.29 bar hits a bite, stands | 1 · 1 · 1 · 1 bar hits | 18.55 · 18.33 · 20.59 · 23.58 of 65 · 65 · 71 · 77 (0.9 × a rifleman HP at his rank, whole) | 0.5 bar hit + thrown 156–158 px a pass | 28.54 · 28.19 · 28.4 · 39.29 bar hits a cut |

A hero's share is 1 (his HP is not on the unit sheet), so a glob lands the whole 2 × the damper's 50 % = 1; a lob lands 0.9 × a rifleman's HP through his bar (4–14 lobs KO a hero at these levels). A priced bite (the pounce, the landing, the sword, Jane's fall under a moving horse) takes 36 HP (49.5 on MEGA) through his bar ÷ his HP: every hero stands. The other heroes take 0.5 bar hit from the hooves and are thrown a pass. The cavalry never take a hero's last blow; **the Emperor's hooves do**: a hero one blow from the KO falls under his charge (measured KING REX on every tier).

## The level a man first survives

For every hit since CARD-BDC2 (no hit is an instant KO). A priced bite from a full bar never fells a man other than a rifleman alone (CARD-RZB's floor), so every one of them lives through one at any rank; the rifleman from the rank his HP passes 36. **Green** = he already does at the lab's EASY level 50; — = not even at rank 100. A glob lands at 50 % (the damper); a lob and a bite land whole.

| | | | GLOBS IN A ROW (Snugg · paratrooper) | | | | CANNON LOBS IN A ROW | | | PRICED BITES IN A ROW (36 HP) | | | BLUECOAT CUTS IN A ROW (Sarge) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| man | HP / bar at 50 | HP / bar at 100 | 2 | 4 | 6 | 8 | 1 | 2 | 3 | 1 | 2 | 3 | 8 | 16 | 24 |
| **RIFLEMAN** | 26 / 7 | 45 / 8 | **0** | **5** | **19** | 100 | **0** | — | — | 80 | — | — | · | · | · |
| **M. GUNNER** | 40 / 7 | 65 / 8 | **0** | **5** | **19** | 100 | **0** | **0** | **1** | **0** | — | — | · | · | · |
| **BAZOOKA** | 80 / 7 | 142 / 8 | **0** | **5** | **19** | 100 | **0** | **0** | **0** | **0** | **50** | 80 | · | · | · |
| **MORTAR** | 23 / 7 | 40 / 8 | **0** | **6** | **19** | 100 | **0** | — | — | **0** | — | — | · | · | · |
| **MEDIC** | 22 / 6 | 40 / 7 | **3** | **10** | **42** | — | **0** | — | — | **0** | — | — | · | · | · |
| **FLAME** | 121 / 7 | 170 / 8 | **0** | **5** | **19** | 100 | **0** | **0** | **0** | **0** | **30** | **50** | · | · | · |
| **SWEEPER** | 24 / 7 | 42 / 8 | **0** | **6** | **19** | 100 | **0** | — | — | **0** | — | — | · | · | · |
| **RADIO MAN** | 24 / 7 | 42 / 8 | **0** | **6** | **19** | 100 | **0** | — | — | **0** | — | — | · | · | · |
| **GRENADIER** | 45 / 7 | 84 / 8 | **0** | **6** | **19** | 100 | **0** | 60 | — | **0** | 90 | — | · | · | · |
| **COMMANDO** | 83 / 8 | 152 / 9 | **0** | **0** | **10** | **42** | **0** | **0** | **0** | **0** | **50** | 70 | · | · | · |
| **SPY** | 27 / 6 | 50 / 7 | **3** | **10** | **42** | — | **0** | — | — | **0** | — | — | · | · | · |
| **SARGE · SABER** | 35 / 12 | 50 / 13 | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | — | — | **0** | **0** | **42** |
| **SARGE · PISTOL** | 35 / 11 | 50 / 12 | **0** | **0** | **10** | **42** | **0** | **0** | **1** | **0** | — | — | **0** | **42** | 100 |
| **SARGE · SHOTGUN** | 135 / 8 | 250 / 9 | **0** | **0** | **10** | **42** | **0** | **0** | **0** | **0** | **30** | **40** | **0** | **42** | — |

| | | | GLOBS IN A ROW (Snugg · paratrooper) | | | | CANNON LOBS IN A ROW | | | PRICED BITES IN A ROW (36 HP) | | | BLUECOAT CUTS IN A ROW (Sarge) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| man | HP / bar at 50 | HP / bar at 100 | 2 | 4 | 6 | 8 | 1 | 2 | 3 | 1 | 2 | 3 | 8 | 16 | 24 |
| **SARGE · SNIPER** | 26 / 8 | 40 / 9 | 0 | 0 | 10 | 42 | 0 | 0 | — | 0 | — | — | 0 | 42 | — |
| **SARGE · CLAYMORE** | 68 / 23 | 78 / 24 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 71 | — | 0 | 0 | 0 |

Read from the game's own HP and bar ladders at every rank 0–100 (the tenth-rank extras and the heavy class included; no gifts, soul or gear), by the sum of the hits with no mend between. Sarge's rows read the iron at the same rank. A lob reads a rifleman's HP at the man's own rank (CARD-EBA): ten lobs in a row, no rank of any man or iron survives. The priced bites read EASY · MEDIUM · HARD's 36 HP (MEGA bites 49.5).

## The units

| unit | attack | what a hit does | HP EASY | MED | HARD | MEGA | n EASY | MED | HARD | MEGA | leak | tags |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
|  **THE HONOR GUARD (bearskins)**<br>walks 40 px/s · 8 HP × the tier | **— (no weapon)**<br>four walk on 1.6 s in and take their seats behind the Emperor (measured: no hit on a rifleman in 5 s) | none: he escorts the Emperor and dives in front of every round meant for him | 8 | 11 | 12 | 14 | 4 | 4 | 4 | 4 | 2 | 20 |
|  **THE SNUGG**<br>walks 45 px/s, rushes ×2.2 with nobody in reach | **GLOB**<br>a glob every 3.6 s at a man within 280 px (measured gaps 3.58–4.17 s); a man left alone 4 s mends 1 bar hit | **2 bar hits** × the damper's 50 % × 90 % + 10 % × bar ÷ HP · on a level-50/57/63/70 rifleman 0.93 · 0.93 · 0.92 · 0.92 · the saber Sarge turns half | 5 | 7 | 8 | 9 | 6 | 8 | 9 | 9 | 1 | 12 |
|  **THE PARATROOPER**<br>walks 40 px/s, rushes ×2 · cannot be aimed at while he hangs | **THE DROP'S THUMP · GLOB**<br>drops 3 s onto the deep band (x 75–85 % of the mat), tangles 0.6 s, then a glob every 2.6 s within 230 px | **a priced bite** of 36 HP (49.5 on MEGA) (no instant KO, CARD-BDC2; from a full bar a man other than a rifleman stands on 15 %, CARD-RZB; one landing reaches at most 3 · 4 · 4 · 5 men) to every man within 76 px of where he lands (60 + 30 % of the man's width; the landing point sways up to 9 px under the chute): measured on eight men per tier and engine, 0 px was reached on every run; 50 px was reached on every run; 80 px was reached on 2 of 8 runs and stood clear on the rest; 100 px stood clear on every run; 130 px and beyond stood clear · then **2 bar hits** × the damper's 50 % × the share a glob | 12 | 16 | 18 | 22 | 2 | 3 | 3 | 3 | 1 | 21 |
|  **THE CUB**<br>runs 140 px/s · 0.75 HP × the tier (rounds to 1) | **POUNCE**<br>hops 460 px/s at a man within 150 px, strikes at contact, the next man 0.9 s later, no cap | **a priced bite** of 36 HP (49.5 on MEGA): a level-50/57/63/70 rifleman falls; every other man stands (from a full bar on 15 % at least, the flame takes 2.08 of 7 bar hits) · the saber Sarge turns about 3 in 4 (his blade rolls on the strike and again on the bite) | 1 | 1 | 1 | 1 | 6 | 6 | 6 | 6 | 1 | 6 |

| unit | attack | what a hit does | HP EASY | HP MED | HP HARD | HP MEGA | n EASY | n MED | n HARD | n MEGA | leak | tags |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
|  **THE CAVALRY (three, at the keel)** 105 HP × 0.6 × the tier × the Emperor's chart multiplier: his bar rides the Emperor's (the bars table) · leaks 6 lives at the box | **THE HOOVES** prances 1.2 s, then charges 420 px/s through the thickest cluster every 2nd leg (rides 180 px/s between); a man within 52 px is trampled, once every 0.9 s each | **a priced bite** of 36 HP (49.5 on MEGA) to every man he rides over · Sarge's and Jane's fall under a moving horse is priced too (a saber Sarge's blade may turn the bite) · another hero: **0.5 bar hit** (1 × the damper) + thrown 89–219 px a pass (King Croak 23–29; measured) · the jeep and the tank ride through untouched | 514 | 605 | 820 | 1845 | 3 | 3 | 3 | 3 | 6 | 150 |
|  **THE BLUECOAT (the Emperor's funnel)** walks 64 px/s, rushes ×1.5 · 8 HP × the tier × 1.5 × THE PRICE of the rung (the HP columns = solo at the lab's level; a party, or rung 50 or under: 12 · 17 · 18 · 21, measured) · comes in details of 2 (a battalion of 4 + a leader every third), a column behind the officer | **DASH-SLASH** a man within 330 px: a wind-up, a flight at 680 px/s to a pace short, three cuts 0.22 s apart at 74 px; then a 6 s clock | **a priced bite** of 36 HP (49.5 on MEGA) to a man (on a Flamer 3.6 HP, CARD-EBA: 34 cuts on a level-50 flame) · **0.5 bar hit** to Sarge or a hero (1 × the damper's 50 %, × the share on Sarge; the saber turns half) · never a glob: he fights with the blade | 12 | 19 | 23 | 30 | ∞ | ∞ | ∞ | ∞ | 2 | 16 |
|  **THE BLUECOAT OFFICER** 24 HP × the tier × THE PRICE of the rung (solo at the lab's level; a party: 24 · 32 · 36 · 43, measured) · walks 64 px/s | **DASH-SLASH · THE WHISTLE** one on the field at a time; he rides in with the first detail and every detail that finds none standing; his fall breaks the line | the same cuts as a Bluecoat · his whistle hups every Bluecoat within 300 px and marches them ×1.3 for 4 s | 24 | 36 | 45 | 61 | ∞ | ∞ | ∞ | ∞ | 2 | 40 |
|  **THE COTTON CANNON (the Emperor's battery)** rolls 40 px/s west and parks at x 1130 behind the river · 10 HP × the tier × 2.5 × THE PRICE of the rung (solo at the lab's level; a party: 25 · 35 · 38 · 45, measured) · two at the ride's start, more with every wave | **COTTON LOB** a lob every 3 s (+ a 0.5 s wind) at the nearest man within 1100 px, the first 85 % primed; lands within 40 px of him, bursts 50 px wide, flies 0.6–1.4 s | **0.9 × a rifleman's HP** at the man's rank through his bar a lob, landing whole (CARD-EBA; a rifleman 0.9 of his bar; a bar it would empty from full leaves 15 %) (every man: 2–6 lobs to KO at the lab's levels; a hero: 4–14) · the landing hurts the bears in its splash too (23.4 HP, 33.3 on MEGA: a Bluecoat a fifth of it, the Emperor two fifths) · the jeep and the tank: one hull hit a lob | 25 | 40 | 47 | 64 | ∞ | ∞ | ∞ | ∞ | 2 | 24 |

| unit | attack | what a hit does | HP EASY | MED | HARD | MEGA | n EASY | MED | HARD | MEGA | leak | tags |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
|  **EMPEROR BEAR** rides 140 px/s, charges 420; walks 190 px/s on foot · leaks 8 lives | **see Emperor Bear below** above 80 % of his first bar the sword from the saddle (one man every 1.5 s); under it the hooves and the charge, the saddle swing at rest; on foot the sword's three cuts and his parry; two bars | **a priced bite** of 36 HP (49.5 on MEGA) every blow on a man; a hero 1 bar hit × the damper a pass of the hooves | 1240 | 1653 | 2478 | 6339 | 1 | 1 | 1 | 1 | 8 | 250 |

HP = each body as the fields' own wave groups bear it on that tier (measured in this read: the plush are the tier alone). The Bluecoats, the officer and the cannons come through the Emperor's funnel at ×1.5 (Bluecoat) and ×2.5 (cannon) **× the price of the player's rung past 50** (the funnel's bodies keep the price the Emperor's bar no longer carries): the HP columns are the lab's SOLO bodies at level 50 · 57 · 63 · 70 (×1 · 1.136 · 1.257 · 1.422); a party, or a player at rung 50 or under, meets a Bluecoat of 12 · 17 · 18 · 21, an officer of 24 · 32 · 36 · 43 and a cannon of 25 · 35 · 38 · 45 HP (measured on both engines); at rung 100 solo the price is ×2.167. n = bodies named in the solo wave sheet (one commander); ∞ = as many as the funnel brings while he lives. leak = lives it costs at the toy box. tags = dog tags paid when it falls.

## Emperor Bear, move by move

| move | EASY | MEDIUM | HARD | MEGA | how it works |
|---|---|---|---|---|---|
| **THE RIDE** | 140 px/s | 140 px/s | 140 px/s | 140 px/s | He walks on at the east rim and mounts at once (no court, no speech): the battery of 2 cotton cannons is born a screen and a half in (x 1500), the first pyramid bursts, the horn. **At or above 80 % of his first bar he walks and cuts from the saddle** (the next row, CARD-EBP); under it the prance and the horn, then every leg he rides AT a man picked at random east of x 300 (never the toy box), a horse-length short of him, re-aimed as the man moves. |
| **THE SWORD FROM THE SADDLE** | **36 HP** a cut | **36 HP** a cut | **36 HP** a cut | **49.5 HP** a cut | Above 80 % of his first bar (CARD-EBP, Nov 10) he walks his horse at 70 px/s at the NEAREST of your men east of x 300, pulls up 80 px off and cuts him from the saddle every 1.5 s inside 110 px: one man a swing, the priced bite (no trample while he walks; the guard, the battery and the funnel go on). Measured on every tier, both engines: a rifleman 300 px off falls to the first cut (9.69 of his 7 bar hits); a flame 300 px off is in reach after 2.72 s (2.71 by the walk) and takes 2.08 of his 7 a cut, 1.52 s apart. |
| **THE HOOVES** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | Under 80 % of his first bar every man within 52 px of the moving horse is trampled, once every 0.6 s each: **a priced bite** (no instant KO since CARD-BDC2: 36 HP (49.5 on MEGA), landing whole; from a full bar a man other than a rifleman stands on 15 %, CARD-RZB). **Sarge (every class) and Jane**: the fall under a moving horse comes already rolled and is priced too (a saber Sarge's blade turns the bite in its own roll). Another hero takes 0.5 bar hit (1 × the damper's 50 %) and is thrown 202–208 px a pass (measured on both engines, every tier; EASY on Chrome: the rifleman falls · the flame stands (2.08 of 7 bar hits) · the bazooka stands (3.15 of 7 bar hits) · the medic stands on 15 % · Sarge falls · Jane stands (18.67 of 56 bar hits) · King Rex 0.5 bar hit + thrown 203 px · the machine gunner stands (6.3 of 7 bar hits), a later pass fells him), **and under the Emperor every pass counts**: a hero one blow from the KO falls (measured KING REX at 224.5 of 225 bar hits: KO under the Emperor's charge on every tier; under a cavalryman's charge he stands untouched, thrown 159 px: the cavalry never take a hero's last). The jeep and the tank ride through untouched. |
| **THE CHARGE** | 420 px/s | 420 px/s | 420 px/s | 420 px/s | Every 2nd–3rd leg (every 1st–2nd when the rung's price stamped on a solo fight is ×1.5 or more: the price no longer moves his bar, but it still sets his temper and his army's bodies): he rears 0.9 s on the spot (the dust ring, the horn, a red ring on his quarry), then runs the man down at 420 px/s and 120 px past him, the hooves landing on everyone in the way. At 70 % and 35 % of his bar the next leg is a charge now. |
| **THE SADDLE SWING** | **36 HP** | **36 HP** | **36 HP** | **49.5 HP** | At rest between legs (under 80 % of his first bar) he swings at whoever stands within 52 px: a man takes the priced bite (measured: a rifleman 40 px off falls on every tier). **A Sarge with a blade in hand** (the saber, the Claymore) takes 1 bar hit × the damper's 50 % × his share and keeps his feet (measured: the saber Sarge 0.47 bar hit, moved 11 px; his block turned the swing on 1 of 8 runs). **A gun Sarge and every hero take the same blow AND the sideways throw** (measured: the pistol Sarge 0.47 bar hit, thrown 184 px; KING REX 0.5 bar hit, thrown 135 px; on every tier, both engines). |

| move | EASY | MEDIUM | HARD | MEGA | how it works |
|---|---|---|---|---|---|
| **THE ROYAL FUNNEL** | every 9–14 s | every 9–14 s | every 9–14 s | every 9–14 s | While he rides, 5 s in and then every 9–14 s a detail of 2 Bluecoats walks on from the east rim on his lane (measured 12 · 19 · 23 · 30 HP each at the lab's four levels: the Bluecoat's body ×1.5 × the price of the rung; a party meets 12 · 17 · 18 · 21), in a column behind THE OFFICER (24 · 36 · 45 · 61 HP; a party 24 · 32 · 36 · 43), who comes with the first detail and whenever none stands. Every third detail is a battalion: 4 Bluecoats behind a flagged leader, and a fresh cannon if his battery is short (measured). The next detail waits while more than 6 Bluecoats stand. |
| **THE KEEL · THREE SECONDS** | life 1 → 2 | life 1 → 2 | life 1 → 2 | life 1 → 2 | At 0 on the horse he is not felled (**measured through the game's own door on every tier**): the rocker keels and he rises on foot with a **second full bar**, the same number as the first (1240 · 1653 · 2478 · 6339), and the second pyramid bursts. **Since CARD-EBP the unhorse takes 3 s**: nothing lands on him until the whistle (measured 3.02 s); he scampers to the back of his side (x 1750, measured), says **"Suivez-moi, à la gloire!"** in his window at 2 s (EN GARDE! is no longer said), blows the whistle at 3 s, and every incoming wave waits for it. You get **150 dog tags** at the unhorse (measured). The EN GARDE! wave walks on at the whistle (released 1.22 s after it, measured): 4 · 5 · 6 · 6 Snuggs, 3 cubs and the 3 cavalrymen (bar 514 each on EASY). |
| **THE HUNT ON FOOT** | 190 px/s | 190 px/s | 190 px/s | 190 px/s | He runs at a man picked at random east of x 300 (never west of it, never over the water: he crosses at the planks), stops at the sword's reach (74 px) and strikes; a man who falls or leaves is re-picked at once. Measured: a rifleman 150 px off is reached and felled by the priced bite in 0.5 s. |
| **THE LUNGE · THREE CUTS** | **36 HP** × 3 men | **36 HP** × 3 men | **36 HP** × 3 men | **49.5 HP** × 3 men | A pick within 330 px on his bank: a 0.05 s wind-up (the sabre back over his shoulder, a steel ring at his feet), a flight at 680 px/s to a pace short of him, then **three cuts 0.22 s apart**, each the nearest man not yet cut within 74 px in front: **a priced bite each** (no instant KO since CARD-BDC2; measured, three men in reach on EASY: the flame stands (2.08 of 7 bar hits) · the medic stands on 15 % · the rifleman falls; a rifleman 300 px off felled 0.683 s after the clock was ready). The clock is 4.5 s; once in reach he strikes 0.8 s into it. A saber Sarge in reach: the fall comes already rolled and priced, his blade turns the bite in its own roll (154 cuts turned in 80 exchanges); the first cut that lands leaves him standing (on 15 % on EASY, on 15 % on MEDIUM, on 0.32 bar hits on HARD, on 15 % on MEGA) and the next one fells him, never the first (measured 0 of 80). **HIS PARRY** (phase 2, CARD-EBP): 20 % of a saber or Claymore Sarge's swings are blocked and stall him 0.9 s, never twice running (measured 83 of 453 swings). Every hero takes the same bite a cut and stands (measured, all six on EASY: KING YETI 34.81 · KING REX 27.36 · KING CROAK 31.96 · JANE 18.67 · EMPEROR BEAR (the hero) 26.64 · PRINCESS CROAK 28.54 bar hits). |
| **MEN WITHOUT FEAR** | every 8–15 s | every 8–15 s | every 8–15 s | every 8–15 s | While he stands, every 8–15 s a wave of 3 Bluecoats (at most 12 on the field) and 2 cotton cannons walks on from the east (the cannons born at x 1660); the next wave waits while more than 4 of the last stand. At 75 % of his second bar a wave comes now. **THE FINAL PHASE** (the second bar at or under half): every 6–10 s and no waiting (measured). |
| **THE GATES** | 80 · 70 · 35 % / 75 · 50 · 25 % | 80 · 70 · 35 % / 75 · 50 · 25 % | 80 · 70 · 35 % / 75 · 50 · 25 % | 80 · 70 · 35 % / 75 · 50 · 25 % | On the horse: at 80 % he leaves the saddle walk for the legs and the charge (CARD-EBP), at 70 % and 35 % a charge now. On foot: at 75 % a wave now, at 50 % the stone wall (OFF today), at 25 % a lunge now. |
| **THE STONE WALL · THE REFLECT** | off | off | off | off | Both switches are off in the client today (the stone wall off, the reflect off): he neither braces against your rounds nor catches rockets and mortar bombs. |
| **HIS BARS (one commander, the lab's level)** | **1,240** twice | **1,653** twice | **2,478** twice | **6,339** twice | The designers' time-to-kill chart's number for one commander on EASY, MEDIUM and HARD (1,240 · 1,654 · 2,481 at rung 50 or under) and the player-damage-threshold pass's table on MEGA (6,345), each landed at the lab's level 0 to 6 HP under it: the pass writes the bar into his row's multiplier at four decimals over the priced solo bar, so the price of the rung (below) cancels — two full bars, one after the other. With more commanders both grow (the table below). His fall ends the fight; the Bluecoats and the cannons run for the rim, the plush go home. |
| **THE PRICE OF THE RUNG** | ×1 | ×1.136 | ×1.257 | ×1.422 | Solo, a player past campaign rung 50 is priced like a boss of his own rung (the lab's stock army is level 50 · 57 · 63 · 70, so the lab pays ×1 · 1.136 · 1.257 · 1.422); the ladder: rung 50 ×1 · rung 60 ×1.196 · rung 70 ×1.422 · rung 80 ×1.678 · rung 90 ×1.937 · rung 100 ×2.167, capped ×4. **Since the player-damage-threshold pass (v29.30) the price no longer moves the Emperor's bar**: his multiplier is figured over the priced bar, so it cancels (1,653 at level 57 against 1,654 at rung 50 or under on MEDIUM). **It still rides his Bluecoats, his officer and his cannons** (the units table), **and it LOWERS a cavalryman's bar**, the same multiplier over the Emperor's priced bar (605 at level 57, 687 at rung 50 or under on MEDIUM). **A party pays the tier alone** (the table below). |
| **THE TOY BOX** | –8 lives if he reaches it | –8 lives if he reaches it | –8 lives if he reaches it | –8 lives if he reaches it | He never marches the box: his legs run at your men. A cavalryman leaks 6, a Bluecoat, an officer, a guard or a cannon 2, a Snugg, a paratrooper or a cub 1. |

## His bars by the number of commanders (the Emperor, each of his two bars · a cavalryman)

| | 1 commander | 2 commanders | 3 commanders | 4 commanders | 5 commanders | 10 commanders |
|---|---|---|---|---|---|---|
| **EASY**<br>a real party (the tier alone) | 1,240 · cavalry 514 | 2,358[m] · cavalry 979 | 3,475[m] · cavalry 1,439 | 4,585 · cavalry 1,904 | 5,702[m] · cavalry 2,364 | 11,282 · cavalry 4,679 |
| EASY · THE BOSS LAB's party slider (a solo fight with a party's bar, its multiplier figured through the price ×1: the same as a real party's: the price is ×1) | 1,240 | 2,358 | 3,475 | 4,585 | 5,702 | 11,282 |
| **MEDIUM**<br>a real party (the tier alone) | 1,653 · cavalry 605 | 3,147[m] · cavalry 1,302 | 4,639[m] · cavalry 1,922 | 6,124 · cavalry 2,537 | 7,616[m] · cavalry 3,156 | 15,064 · cavalry 6,245 |
| MEDIUM · THE BOSS LAB's party slider (a solo fight with a party's bar, its multiplier figured through the price ×1.136: 2–8 HP under a real party's) | 1,653 | 3,145 | 4,637 | 6,121 | 7,613 | 15,056 |
| **HARD**<br>a real party (the tier alone) | 2,478 · cavalry 820 | 4,712[m] · cavalry 1,952 | 6,942[m] · cavalry 2,879 | 9,184 · cavalry 3,806 | 11,415[m] · cavalry 4,733 | 22,579 · cavalry 9,356 |
| HARD · THE BOSS LAB's party slider (a solo fight with a party's bar, its multiplier figured through the price ×1.257: 7–31 HP under a real party's) | 2,478 | 4,705 | 6,933 | 9,171 | 11,399 | 22,548 |
| **MEGA**<br>a real party (the tier alone) | 6,339 · cavalry 1,845 | 12,042[m] · cavalry 4,988 | 17,738[m] · cavalry 7,350 | 23,435 · cavalry 9,712 | 29,155[m] · cavalry 12,074 | 57,661 · cavalry 23,898 |
| MEGA · THE BOSS LAB's party slider (a solo fight with a party's bar, its multiplier figured through the price ×1.422: 12–55 HP under a real party's) | 6,339 | 12,030 | 17,721 | 23,412 | 29,127 | 57,606 |

**A real party pays the tier alone**: the price of the rung is ×1 while a party launches (the game's twin-sim law: every commander's sim builds the same army from the tier), so from 2 commanders up each of his two bars is the tier's base grown 1 + 0.9 a commander past the first under the chart's multiplier, read from the game's own sheet with the party flag set (2,358 · 3,147 · 4,712 · 12,042 with two); [m] = the Emperor born in a party of that size was measured at exactly this bar on every tier, both engines. One commander is a solo fight: his bar is the chart's number as the game lands it at the lab's level (0 to 6 HP under it), the price of that level cancelled inside the multiplier. THE BOSS LAB's own party slider (the second line) is a solo fight with a party's bar, its multiplier figured through the price, so from MEDIUM up it reads 2–55 HP under what a real party meets (it read higher before the player-damage-threshold pass (v29.30)). Each of his two bars is this number. A cavalryman's bar rides the Emperor's multiplier: a solo player's price lowers it (514 · 687 · 1,032 · 2,626 at rung 50 or under), a party's rows carry none (979 · 1,302 · 1,952 · 4,988 with two).

## The cavalry's hooves on a line of eight, measured (DUKE BEAR's brain: the prance, then the charge through the line)

| | a cavalryman's bar | what each took |
|---|---|---|
| **EASY** | 514 | the rifleman falls · the flame stands (2.08 of 7 bar hits) · the bazooka stands (3.15 of 7 bar hits) · the medic stands on 15 % · the machine gunner stands (6.3 of 7 bar hits) · Sarge stands on 15 %, a later pass fells him · Jane stands (18.67 of 56 bar hits) · King Rex 0.5 bar hit + thrown 215 px |
| **MEDIUM** | 605 | the rifleman falls · the flame stands (2.08 of 7 bar hits) · the bazooka stands (3.15 of 7 bar hits) · the medic stands on 15 % · the machine gunner stands (6.3 of 7 bar hits) · Sarge his blade turned the bite · Jane stands (18.16 of 56 bar hits) · King Rex 0.5 bar hit + thrown 215 px |
| **HARD** | 820 | the rifleman falls · the flame stands (1.94 of 7 bar hits) · the bazooka stands (2.8 of 7 bar hits) · the medic stands on 15 % · the machine gunner stands (5.6 of 7 bar hits) · Sarge stands (11.68 of 12 bar hits), a later pass fells him · Jane stands (18.92 of 62 bar hits) · King Rex 0.5 bar hit + thrown 215 px |
| **MEGA** | 1845 | the rifleman falls · the flame stands (2.48 of 7 bar hits) · the bazooka stands (3.43 of 7 bar hits) · the medic stands on 15 % · the machine gunner stands (6.93 of 7 bar hits) · Sarge his blade turned the bite · Jane stands (26.71 of 68 bar hits) · King Rex 0.5 bar hit + thrown 219 px |

## The funnel, measured

| | what walked on |
|---|---|
| **EASY** | battery 2 × 25 HP at x 1500 · first detail 2 Bluecoats 12 HP + the officer 24 HP · the battalion 5 + 1 cannon · on foot: 3 Bluecoats + 2 cannons (25 HP), the final phase next in 9.09 s |
| **MEDIUM** | battery 2 × 40 HP at x 1500 · first detail 2 Bluecoats 19 HP + the officer 36 HP · the battalion 5 + 1 cannon · on foot: 3 Bluecoats + 2 cannons (40 HP), the final phase next in 6.5 s |
| **HARD** | battery 2 × 47 HP at x 1500 · first detail 2 Bluecoats 23 HP + the officer 45 HP · the battalion 5 + 1 cannon · on foot: 3 Bluecoats + 2 cannons (47 HP), the final phase next in 8.24 s |
| **MEGA** | battery 2 × 64 HP at x 1500 · first detail 2 Bluecoats 30 HP + the officer 61 HP · the battalion 5 + 1 cannon · on foot: 3 Bluecoats + 2 cannons (64 HP), the final phase next in 6.31 s |

## What changes from EASY to MEGA

| | EASY | MEDIUM | HARD | MEGA |
|---|---|---|---|---|
| **The priced bite (a former KO, CARD-BDC2: the fight's level 100, MEGA 140; landing whole)** | 36 HP | 36 HP | 36 HP | 49.5 HP |
| **One blow reaches at most (CARD-RZB)** | 3 men | 4 men | 4 men | 5 men |
| **Every other hit (a glob 2 bar hits landing at 50 % by the damper; a lob 0.9 × a rifleman's HP, whole)** | | | the same on every tier | |
| **Body HP (the plush: the tier alone)** | ×1 | ×1.35 | ×1.5 | ×1.8 |
| **The price of the rung on his Bluecoats, officer and cannons (solo at the lab's level; ×1 in a party; since the player-damage-threshold pass (v29.30) it no longer moves the Emperor's bar)** | ×1 | ×1.136 | ×1.257 | ×1.422 |
| **The Emperor's two bars at the lab's level (solo)** | 1,240 | 1,653 | 2,478 | 6,339 |
| **… at rung 50 or under: the designers' time-to-kill chart's own number (MEGA the player-damage-threshold pass's table)** | 1,240 | 1,654 | 2,481 | 6,345 |
| **The Emperor's two bars with 2 commanders (a real party: the tier alone)** | 2,358 | 3,147 | 4,712 | 12,042 |
| **A Bluecoat · the officer · a cannon, solo at the lab's level** | 12 · 24 · 25 | 19 · 36 · 40 | 23 · 45 · 47 | 30 · 61 · 64 |
| **… in a party, or at rung 50 or under (measured)** | 12 · 24 · 25 | 17 · 32 · 35 | 18 · 36 · 38 | 21 · 43 · 45 |
| **A cavalryman's bar, solo at the lab's level (it rides the Emperor's multiplier)** | 514 | 605 | 820 | 1,845 |
| **… at rung 50 or under (a solo player's price LOWERS it; a party's rows carry none)** | 514 | 687 | 1,032 | 2,626 |
| **Snuggs (the opening + EN GARDE!)** | 2 + 4 | 3 + 5 | 3 + 6 | 3 + 6 |
| **Paratroopers** | 2 | 3 | 3 | 3 |
| **Guards · cubs · cavalry** | | | 4 · 3 + 3 · 3 | |
| **Bodies in the sheet, solo (the Emperor and the cavalry not counted)** | 18 | 21 | 22 | 22 |
| **The lab's stock army** | level 50 | level 57 | level 63 | level 70 |
| **The rung's price at that level (solo: on his Bluecoats, officer and cannons; it lowers the cavalry's bar; it no longer moves the Emperor's)** | ×1 | ×1.136 | ×1.257 | ×1.422 |

| | EASY | MEDIUM | HARD | MEGA |
|---|---|---|---|---|
| **MEGA's door** | — | — | — | the whole 50-rung Bear Park ladder climbed |
| **The castle's +20 % per hit and double weather** | | | not here: that is the Ice Castle's own law | |

## The wave sheets (solo, one commander)

**EASY · 18 bodies + the funnel**

1. **EMPEROR BEAR TAKES THE FIELD** (ON THE CLOCK)
   4 honor guards, 2 Snuggs, 2 paratroopers, 3 cubs · then EMPEROR BEAR (bar 1,240 × 2) at 3 s
2. **EN GARDE!** (HOLDS WHILE THE EMPEROR IS ON HIS HORSE)
   4 Snuggs, 3 cubs, 3 CAVALRYMEN

**MEDIUM · 21 bodies + the funnel**

1. **EMPEROR BEAR TAKES THE FIELD** (ON THE CLOCK)
   4 honor guards, 3 Snuggs, 3 paratroopers, 3 cubs · then EMPEROR BEAR (bar 1,653 × 2) at 3 s
2. **EN GARDE!** (HOLDS WHILE THE EMPEROR IS ON HIS HORSE)
   5 Snuggs, 3 cubs, 3 CAVALRYMEN

**HARD · 22 bodies + the funnel**

1. **EMPEROR BEAR TAKES THE FIELD** (ON THE CLOCK)
   4 honor guards, 3 Snuggs, 3 paratroopers, 3 cubs · then EMPEROR BEAR (bar 2,478 × 2) at 3 s
2. **EN GARDE!** (HOLDS WHILE THE EMPEROR IS ON HIS HORSE)
   6 Snuggs, 3 cubs, 3 CAVALRYMEN

**MEGA · 22 bodies + the funnel**

1. **EMPEROR BEAR TAKES THE FIELD** (ON THE CLOCK)
   4 honor guards, 3 Snuggs, 3 paratroopers, 3 cubs · then EMPEROR BEAR (bar 6,339 × 2) at 3 s
2. **EN GARDE!** (HOLDS WHILE THE EMPEROR IS ON HIS HORSE)
   6 Snuggs, 3 cubs, 3 CAVALRYMEN

The opening runs on the clock (the guards 1.6 s in, the Emperor at 3.0 s, the Snuggs 4.6 s, the paratroopers 5.4 s, the cubs 5.8 s). EN GARDE! waits for the keel: the Emperor's first bar at 0. On top of both: the battery at the ride's start, a Bluecoat detail every 9–14 s while he rides, a wave of Bluecoats and cannons every 8–15 s while he stands.

## How this was read and checked

The live client v32.45 (index.html md5 3a46a5281956aeaf75ac23e3b689a4ba) was opened headless on this Mac and read with the game's own functions: every table, every tier's wave sheet at the lab's own level, every man's HP and bar at every rank, the Emperor's and the cavalry's bars for 1 to 10 commanders, the price ladder. Then each hit was **measured on the real machines** in a BOSS LAB fight on every tier with the stock army worn (level 50 · 57 · 63 · 70), every plush born through the fields' own wave rows and the Emperor's own doors (the ride, the keel, the funnel, the battery): the cub's pounce on seven men (and 40 on the saber Sarge for his block), an honor guard beside a rifleman, a Snugg's and a paratrooper's glob to the KO, a Bluecoat's and the officer's dash-slash on men and on the saber Sarge (10 and 6 tries), the officer's whistle, the paratrooper's landing on eight men at 0–250 px (the reach 76 px from the landing point), a king's cannon on seven men and the jeep, the cavalry's charge through a line of eight, the Emperor's charge through the same line, his saddle swing on a rifleman, a saber Sarge, a pistol Sarge and King Rex, his battery, his first detail and a battalion, his three cuts on three men, the lunge from 300 px, the walk-in from 150 px, his cut on the saber Sarge (10 tries) and on each hero, MEN WITHOUT FEAR's wave and the final phase, the six heroes against the pounce, the glob, the lob and the hooves, King Rex one blow from the KO under a cavalryman's charge and under the Emperor's, every body as the tier bears it, the funnel's bodies at a party's price, the Emperor born in a party of 2, 3 and 5, and the keel through the game's own door (the snap, the second bar, EN GARDE!, the second pyramid, the EN GARDE! wave's release), with THE DAMPER read in each tier's fight (50 % of every blow lands) and in every landed blow above. Chrome 349/349 · WebKit 349/349 checks passed, zero page errors, and the two engines read every number the same.

Sources (the code's own names): THE v32.45 RE-READ (CARD-BDB-KO): knockKill's price (bdc2Price, bdc2Fight, bdc2Level, bdc2Rank, BDC2_K, BDC2.levels; dsmH's heavy class BDC2_HEAVY), tanHit's floor and knock's cap (rzbBoss, rzbFloorMan, rzbFloorStand, rzbCapped: RZB_FLOOR, RZB_CAP, RZB_WIN), CARD-EBA's wraps (foeSplash, tanHit, knock, bdc2Price, spawnGroup, jlCamp: ebaUnitPips, ebaFriendly, EBA_COT, EBA_FF_TAKE, EBA_CUT_FLAME, EBA_PARA), CARD-EBP's (kbRideStep, kbUnhorse, kbTick, kbCut, hurt: ebpSaddle, ebpUnhorse, ebpSeqTick, ebpBlock; GVT.ebp's dials); RAID_INFO.fields, raidCfg's fields rows (tguard · bearnaparte · tsnugg · tpara · tcub · tcavalry; hpX kbPX · 0.6), FIELDS_HPX, RAID_CNT_MUL, BLB_STOCK_LV, blbStockDef, kbPriceX / kbBossX; FDEF rows; updateFoePounce → knock; tanShoot's pips → tanHit (DSM k 0.9, dsmX / dsmH, gritCap, the 4 s mend); updateFoePara (the THUMP → knock); guardDive; kbRoySeat / kbRoyFight / kbRoyTick / kbRoyForm, kbCut (KB_ROY_CUT_PIP), updateFoeGruffle (the officer); kbCanSeat / cannonTick / foeSplash (KB_COT_F, KB_COT_SCAT); dkStart / dkRideStep / kbTrample (KB_RAM_R, KB_TRAM_CD, KB_RAV_CD, KB_HOOVES); kbRideStart / kbRideStep / kbPickWay / kbGate1 / kbUnhorse / kbTick / kbFootTick / kbHuntFoot / kbWindGo / kbDashGo / kbDashTick / kbCut / kbAnswer / kbGate2 / kbMwfTick / kbBattery (KB_* cells); blbSheet('fields', tier, N), bhpX (BHP_K); THE BAR LAW SINCE 11/2: pdtScale / pdtSolo / pdtRow wrapped on raidCfg (PDT.bars.fields, the row's hpX at four decimals over blbBar for one commander, kbPriceX inside it), rxtApply (RXT.live.fields over PDT.bars on e · m · h), kbArmyPX (the funnel's price), kbRideStart's kbFierce off S.cfg.kbPX; bfHpAt, usBfHP, sgBfHP, heroBfHP, saberBlocks; THE MEN'S TENTH-RANK HP: xhpAt / XHP.per (FLT: the flame's rungs through 50; DHP: bazooka · commando doubled, sgAdd 20 a sgStep 10 of the shotgun's rank); THE DAMPER: DMG_DAMP (e · m · h · g 0.5 · 0.5 · 0.5

· 0.5), dmpTier / dmpShare / dmpX at tanHit's two pip writes (GVT.dmp the read). Kit: design/boss-emperor-unit-breakdown-1031/ (kit/readBear.mjs · kit/proveBear.mjs · kit/build.py · kit/render.mjs · kit/export_dash.py; data/; proof/logs/). Zero game bytes.